

Volleyball Player Clustering

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1 Tasks 1–2: Preparation and cluster selection

Height was parsed to centimetres. Name, team, and position were excluded as required. The clustering variables were age, height, and six per-match activity measures: sets, receives, serves, blocks, digs, and attacks. Rates reduce the effect of unequal playing time, while age and height retain demographic and morphological information. Every variable was standardized to zero mean and unit variance so centimetres, years, and rates contribute on comparable scales. The seed was fixed at 366.

K-means was fitted with 100 random starts for each $k = 2, \dots, 8$. Average silhouette widths were .256, .267, .320, .276, .231, .248, and .254. Thus $k = 4$ gives the strongest tested separation, improving on $k = 3$ and $k = 5$, although .320 indicates overlapping rather than perfectly isolated groups. The final fit used 200 starts.

Cluster	1	2	3	4
Players	30	84	194	31
Share (%)	8.8	24.8	57.2	9.1

Table 1: Size of the selected four-cluster K-means solution.

1.1 K-means interpretation

Cluster 1 has exceptionally high setting activity (18.06 sets per match) and little attacking (0.52), and therefore resembles setters. Cluster 2 is tallest (200.37 cm) and has the highest blocks (1.11) and attacks (8.47), representing attacking/blocking specialists. Cluster 3 is the large lower-to-moderate activity group. Cluster 4 is shortest (183.13 cm) and has the highest receiving (3.95) and digging (4.87), matching a libero profile.

Figure 1 projects the eight-dimensional solution onto the first two principal components, which retain 56.2% of variance. Visual overlap in this projection is therefore not identical to overlap in the complete clustering space.

2 Task 3: Comparison with DBSCAN

DBSCAN used the same standardized variables. Following $\text{minPts} = 2p$, minPts was set to 16; the elbow of the sorted 15-nearest-neighbour distances gave $\epsilon = 2.664$. DBSCAN produced one connected dense cluster containing 333 players and six noise players (1.77% of the sample; Figure 2).

The adjusted Rand index against K-means is only .0282, indicating almost no partition agreement. This does not imply algorithmic failure: K-means forces every observation into compact Voronoi groups, while DBSCAN searches for density-separated regions and may identify noise. Together, the results suggest connected role gradients with a few sparse extremes rather than four isolated density islands.

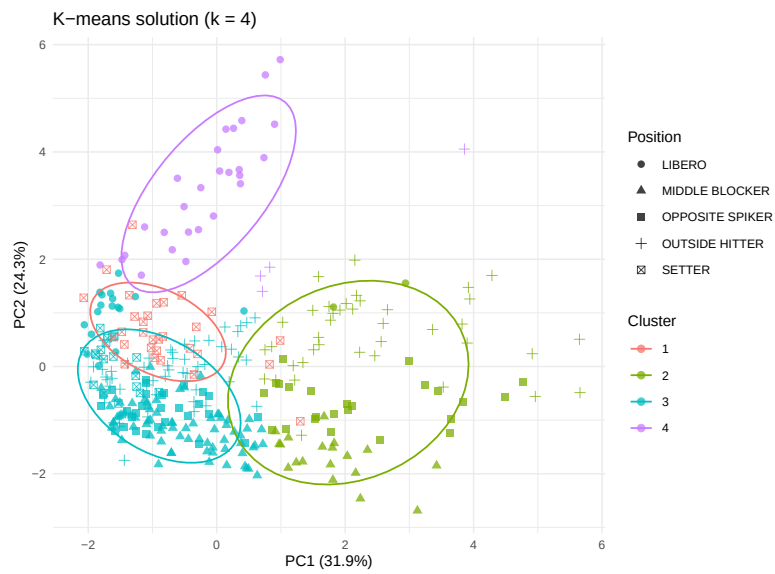


Figure 1: PCA display of the four-cluster K-means solution; position is shown only post hoc.

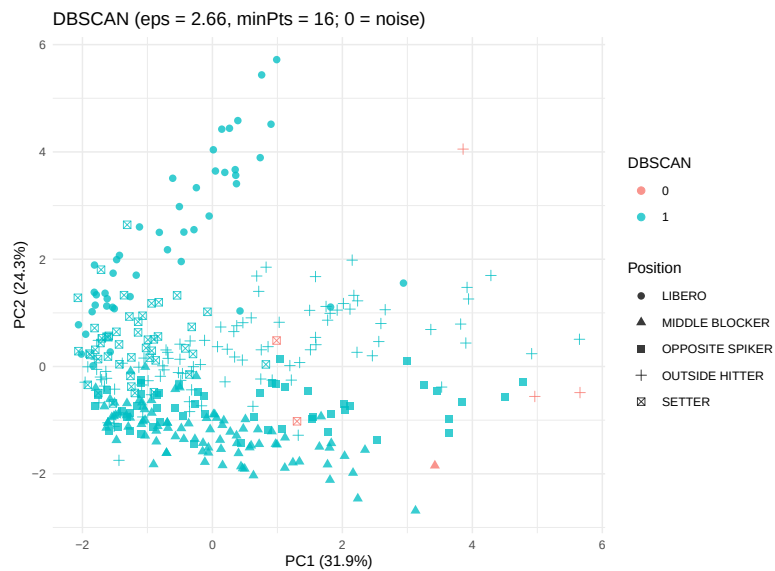


Figure 2: DBSCAN on the same PCA plane; label 0 denotes noise.

3 Task 4: Alignment with position and country

The post-hoc position comparison strongly supports the functional interpretation: Cramér's $V = .607$. Cluster 1 consists entirely of setters; 87.1% of Cluster 4 are liberos; Cluster 2 concentrates hitters and blockers. This is external face validity, not supervised accuracy, because position was excluded from fitting.

Country alignment is weak ($V = .134$), compared with the much stronger position alignment. Each national team necessarily contains several roles and therefore contributes to several clusters. Country distributions may also reflect roster composition and playing opportunity and should not be interpreted as national quality.

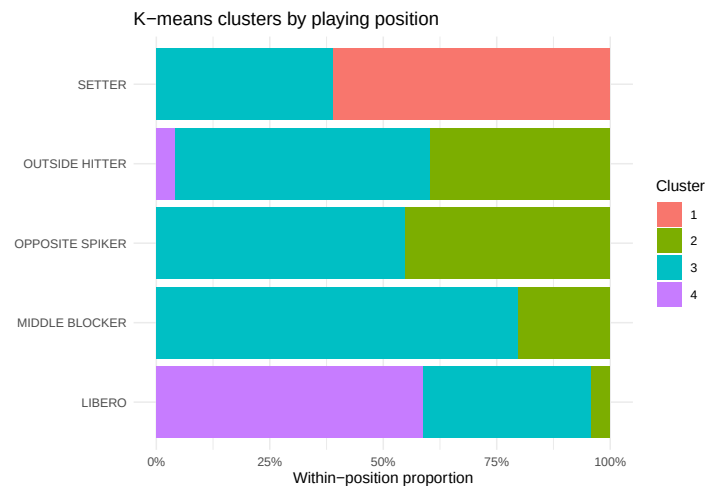


Figure 3: Within-position K-means cluster composition.

4 Conclusion

Four K-means clusters provide the clearest tested compact partition and recover recognisable role profiles. DBSCAN instead supports a single connected population plus six unusual players. Cluster membership aligns strongly with playing position but weakly with country; the groups therefore represent volleyball functions rather than national teams.

Statement of Contributions

Indudhara Swamy Vivekananda completed Exercise 1 (match-statistics preparation, missingness analysis, normality assessment, EDA, visualisation, and interpretation). Sanjana Basoor Hemanth Kumar completed Exercise 2 (player-statistics preparation, descriptive and correlation analysis, outlier analysis, country comparison, and player ranking). Rohit Mamgin completed Exercise 3 (variable selection, K-means and DBSCAN clustering, validation, visualisation, and position/country alignment). All authors reviewed the complete submission and approved the final reports.